CS1231S Cheat Sheet, AY25/26 S2

Zhao Wu

May 08, 2026

1 Proof

1.1 Statements and Logical Equivalence

Statement / Proposition

Statement (or **Proposition**) is a sentence, that is either **True or False, but not both**.

- A sentence with open ended variable (e.g. $x^2 + 2 = 11$) is not a statement as its truth value depends on x.

Compound Statements

Compound statement (or boolean expression) is a statement, constructed from **statement variables** by using **negation** (not), **conjunction** (and) and **disjunction** (or).

- **Negation Statement**, "Not", ($\sim$ /$\neg$)

$$\sim p \text{ is } \begin{cases} \text{true if } p \text{ is false} \\ \text{false if } p \text{ is true} \end{cases}$$

- **Conjunctive Statement**, "And / But / Neither-nor", ($\wedge$ / $\cap$)

$$p \wedge q \text{ is } \begin{cases} \text{true if } p \text{ is true and } q \text{ is true} \\ \text{false if otherwise (either p or q is false)} \end{cases}$$

- **Disjunctive Statement**, "Or / Either-or", ($\vee$ / $\cup$)

$$p \vee q \text{ is } \begin{cases} \text{true if } p \text{ is true or } q \text{ is true} \\ \text{false if otherwise (p and q are false)} \end{cases}$$

Order of operation for Logical Operators

1. $\sim$
2. $\wedge, \vee$ are co-equal in order of operations
3. $\Rightarrow, \Leftrightarrow$ are co-equal in order of operations

Logical Equivalence

Statement P and Q are **logically equivalent ($P \equiv Q$)** $\Leftrightarrow$ they have the same truth values for all substitution of statements for their statement variables.

To show nonequivalence, construct a truth-table with a sentence of explanation where the truth values of both statements are not the same.

Tautologies, t and Contradictions, c

Tautology, t is statement that is **always true** regardless of truth values of individual statement variables.

Contradiction, c is a statement that is **always false** regardless of the truth values of individual statement variables.

1.1.1 Theorem: Logical Equivalences

No.	Law	Equivalences
1	Commutative laws	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
2	Associative laws	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$ $p \vee (q \vee r) \equiv (p \vee q) \vee r$
3	Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4	Identity laws	$p \wedge \boldsymbol{t} \equiv p$ $p \vee \boldsymbol{c} \equiv p$
5	Negation laws	$p \vee \sim p \equiv \boldsymbol{t}$ $p \wedge \sim p \equiv \boldsymbol{c}$
6	Double negative law	$\sim (\sim p) \equiv p$
7	Idempotent laws	$p \wedge p \equiv p$ $p \vee p \equiv p$
8	Universal bound laws	$p \vee \boldsymbol{t} \equiv \boldsymbol{t}$ $p \wedge \boldsymbol{c} \equiv \boldsymbol{c}$
9	De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$ $\sim (p \vee q) \equiv \sim p \wedge \sim q$
10	Absorption laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
11	Negation of true and false	$\sim \boldsymbol{t} \equiv \boldsymbol{c}$ $\sim \boldsymbol{c} \equiv \boldsymbol{t}$

1.2 Conditional Statements

Conditional Statement

Conditional statement is a compound statement that denote the causal relationship between a **hypothesis** p, and **conclusion** q.

$$p \to q; \text{ if p then q; p only if q}$$

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

NOTE: For last 2 rows, since hypothesis p is false, the overall conditional statement $p \to q$ is **vacuously true** (aka **true by default**).

$$p \to q \equiv \sim p \vee q$$
$$\sim (p \to q) \equiv \sim (\sim p \vee q) \text{ (De Morgan's Law)} \equiv p \wedge \sim q$$

- Variations of if/then statements:
  - **If** p **then** q: $p \to q$
  - p **only if** q: $p \to q$
  - p **if** q: $q \to p$
  - p **if and only if** q: $p \leftrightarrow q$

Converse, Inverse, Contrapositive

Given conditional statement $p \to q$, there are a few notable variant:

- **Conditional**, $p \to q \equiv$ **Contrapositive**: $\sim q \to \sim p$
- **Converse**: $q \to p \equiv$ **Inverse**: $\sim p \to \sim q$

Biconditional Statement

Bidirectional Statement, if and only if, is true only when both p and q has the same truth values.

$$p \leftrightarrow q; \text{ p if and only if q}$$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$

Necessary and Sufficient Conditions

r is **Necessary Condition** for s means if not r then not s ($\neg r \to \neg s$)

r is **Sufficient Condition** for s means if r then s ($r \to s$)

1.3 Valid and Invalid Arguments

Definition of Valid Argument, Premise, Conclusion

- **Argument**: sequence of statements.
- **Argument Form**: sequence of statement forms.
- **Premise / Assumptions / Hypothesis**: all statements in an argument except the final one.
- **Conclusion**: final statement in an argument, normally preceded with $\therefore$.
- **Valid Argument**: For all statement variables substituted to its premises, the resulting premises are all true and conclusion is also true. ($\forall x \in \cup, P(x) \wedge C(x)$).
- **Invalid Argument**: Exists at least one premise, where after substituted to its statement variable, its conclusion is not true ($\exists x \in \cup, P(x) \wedge \sim C(x)$).
- **Critical Row**: Row in truth-table where all premises are true but conclusion is false.

Modues Ponens and Modus Tollens

- **Modus ponens** "method of affirming":

$$p \to q \text{ (If its raining, ground is wet)}$$
$$p \text{ (It is raining)}$$
$$\therefore q \text{ (Ground is wet)}$$

- **Modus tollens** "method of denying":

$$p \to q \text{ (If its raining, ground is wet)}$$
$$\sim q \text{ (Ground is not wet)}$$
$$\therefore \sim p \text{ It is not raining}$$

Rule of Inference

1.3.1 Theorem: Valid Argument Form

Rule of Inference			
Modus Ponens	$p \to q$ p $\therefore q$	Modus Tollens	$p \to q$ $\sim q$ $\therefore \sim p$
Generalization	$p \mid q$ $\therefore p \vee q$	Specialization	$p \wedge q$ $\therefore p \mid q$
Conjunction	p q $\therefore p \wedge q$	Elimination	$p \vee q$ $\sim q \mid \sim p$ $\therefore p \mid q$
Transitivity	$p \Rightarrow q$ $q \Rightarrow r$ $\therefore p \Rightarrow r$	Proof by division into cases	$p \vee q$ $p \to r$ $q \to r$ $\therefore r \quad (x^2 > 0)$
Contradiction rule	$\sim p \to$ false $\therefore p$		

Fallacies

- **Converse Error**

$$p \to q$$
$$q$$
$$\therefore p \text{ (Invalid as p may be true or false)}$$

- **Inverse Error**

$$p \to q$$
$$\sim p$$
$$\therefore \sim q \text{ (Invalid as q may be true or false)}$$

1.4 Predicates and Quantification

Predicates

Predicates, (P(x), Q(x,y)) is a sentence with a finite number of predicate variables, that become a statement when its value is substituted.

Domain of predicate variable is set of all values that may be substituted in place of the variable.

Truth set of P(x) is set of all element in domain D where $P(x)$ is true.

$$\{x \in D \mid P(x)\}$$

Universal and Existential Quantifier

- **Universal Statement**:"*for every*", ($\forall x \in S$)
  - **Universal Conditional Statement**:
    - E.g. *For every* animal a, *if* a is a dog, *then* a is a mammal
    - Which can also be written purely conditional or universal:
      - *If* an animal a is a dog, *then* a is a mammal
      - *For every* dog a, a is a mammal
  - **Universal Existential Statement**:
    - E.g. *There exists* additive inverse *for every* real number / *Every* real number *has an* additive inverse
  - **Existential Statement**: "there exist", ($\exists x \in S$)

Equivalent Forms of Universal and Existential Statements

$$\forall x \in U, P(x) \to Q(x)$$
$$\equiv \forall x \in U, \sim Q(x) \to \sim P(x) \text{ (By Contraposition)}$$
$$\equiv \forall x \in D, Q(x) \text{ where } D = \{x \in U \mid P(x)\}$$

Implicit Quantification

- $P(x) \Rightarrow Q(x)$ suggests every element in truth set of P(x) is in truth set of Q(x), $\equiv \forall x, P(x) \to Q(x)$
- $P(x) \Leftrightarrow Q(x)$ suggest P(x) and Q(x) has identical truth sets, $\equiv \forall P(x) \leftrightarrow Q(x)$

Proving and Negation of Quantifier

	True	False
$\forall$	Exhaustive List	Counter-example
$\exists$	Example	Exhaustive List

$$\sim (\forall x \in D, Q(x)) \equiv \exists x \in D, \sim Q(x)$$
$$\sim (\exists x \in D, Q(x)) \equiv \forall x \in D, \sim Q(x)$$

Quantified Rules of Inference

Quantified Rule of Inference	
Universal Instantiation	$\forall x \in D, P(x)$ $a \in D$ $\therefore P(a)$
Universal Generalization	$P(a)$ for every $a \in D$ $\therefore \forall x \in D, P(x)$
Existential Instantiation	$\exists x \in D, P(x)$ $\therefore P(a)$ for some $a \in D$
Existential Generalization	$P(a)$ for some $a \in D$ $\therefore \exists x \in D, P(x)$
Universal Modus Ponens	$\forall x(P(x) \Rightarrow Q(x))$ $P(a)$ for a particular a $\therefore Q(a)$
Universal Modus Tollens	$\forall x(P(x) \Rightarrow Q(x))$ $\sim Q(a)$ for a particular a $\therefore \sim P(a)$
Universal Transitivity	$\forall x(P(x) \Rightarrow Q(x))$ $\forall x(Q(x) \Rightarrow R(x))$ $\therefore \forall x(P(x) \Rightarrow R(x))$

One-to-One and On-To

- **One-to-one** function, $f : X \to Y \Leftrightarrow \forall x_1, x_2, x_1 = x_2 \to f(x_1) = f(x_2)$

(i.e. same input, same output)

- **Onto** function, $f : X \to Y \Leftrightarrow \forall y \in Y, \exists x \in X$ where $y = f(x)$

(i.e. every output has an input)

Well-Ordering Principle

- Let S be nonempty set of integers

- If there exists integer $\mathbb{Z}$ smaller than all integer in S, then there is an integer in S that is smaller or equal to all integers in S.

1.5 Mathematical reasoning

Direct Proof

- **Question**: Proof that Sum of any two even integers is even
- **Formal Restatement** $\forall m \in \mathbb{Z}, n \in \mathbb{Z}\ \text{Even}(m) \wedge \text{Even}(n) \to \text{Even}(m+n)$
- **Starting Point**: Suppose m is even and n is even
- By definition of even number, m = 2j, n = 2k for some integer j, k.
- $$m + n = 2j + 2k$$ $$2(j+k)$$
- Let $t = j + k$, t is integer as it is sum (and products) of integers.
- $\therefore m + n = 2t$ and by definition, m + n is even `[Q.E.D.; quod erat demonstratdum]`

Proof by Cases

- Divide the statement to prove into smaller cases that are more easily provable, then show that the statement is true for each case, hence since the cases cover the complete domain, the statement must be true.

Example

Prove: The only divisors of 1 are 1 and -1

Proof by division into cases:
1. Suppose m is any number that divides 1
   1. Then there exists an integer k such that $1 = mk$ (by definition of divisibility)
   2. Since mk is positive, either both m and k are positive, or both are negative
   3. Case 1: Both m and k are positive.
      1. Since $m \mid 1, m \leq 1$ (by Theorem 4.4.1)
      2. Then $m = 1$
   4. Case 2: Both m and k are negative
      1. Then $-m$ is a positive divisor of 1, i.e. $-m|1$
      2. Then by the same reasoning in 1.3.1 and 1.3.2, $-m = 1$, or $m = -1$
2. Therefore, the only divisors of 1 are 1 and -1

Proof by Exhaustion

- If our domain is finite or when only a finite number of elements satisfy $P(x)$, we may prove the statement by the method of exhaustion (i.e. listing out all possible cases, thus an extension of proof by cases)

Proof by Contraposition

- In proving the contrapositive of the statement, you indirectly prove the statement itself as the two are equivalent.
- If you prove by contraposition, you start with the negation of the conclusion, and end with the negation of the hypothesis.

Example

Prove: Let n be an integer. Show that if n^2 is even, then n is also even

Proof by contraposition:
1. Contrapositive statement: If n is odd, then n^2 is odd
2. Let n be a particular but arbitrarily chosen odd integer
3. Then, $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$ (by definition of odd integers)
4. Then, $n^2 = (2k+1)^2 = 2(2k^2 + 2k) + 1$ (by basic algebra)
5. Note that $2k^2 + 2k \in \mathbb{Z}$ (by closure of integers under addition and multiplication)
6. Therefore, n^2 is an odd integer (by definition of odd integers)
7. Since if n is odd, then n^2 is odd, therefore, if n^2 is even, then n is also even

Proof by Contradiction

- If we want to show that $p \Rightarrow q$ is true, we can also show that $\sim (p \Rightarrow q)$ is false
- Note that $\sim (p \Rightarrow q) \equiv p \wedge \sim q$
- Hence, we start with the antecedent and the negation of the consequent, and we end with any contradiction (i.e. show that $p \wedge \sim q$ is false)

Example

Prove: Suppose x and y are real numbers. If $xy = 0$, then $x = 0$ or $y = 0$

Proof by contradiction:
1. Suppose the claim is false, that is, $\exists x, y \in \mathbb{R}$ such that $xy = 0$ but $x \neq 0$ and $y \neq 0$
2. Since $x \neq 0$, we can divide both sides of $xy = 0$ by x (By Epp, Appendix A, T7)
3. Therefore, $y = 0$ (by basic algebra)
4. But this contradicts $y \neq 0$
5. Therefore, the assumption is false, and the claim is true

Proof by Mathematical Induction (1PI)

- To prove a statement of the form, "For all integers $n \geq a$, a property $P(n)$ is true":
  1. Show that $P(a)$ is true
  2. Show that for all integers $k \geq a$, if $P(k)$ is true, then $P(k+1)$ is true. To do this, suppose that $P(k)$ is true for some arbitrarily chosen $k \geq a$ (called the Inductive Hypothesis), then show that $P(k+1)$ is true

Example

For all integers $n \geq 1$, show $1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$

Proof by Induction:
1. Let $P(n) \equiv \left(1 + 2 + ... + n = \frac{n(n+1)}{2}\right), \forall n \in \mathbb{Z}^+$
2. **Basis Step**:
   - $1 = \frac{1(1+1)}{2}$, therefore $P(1)$ is true
3. **Inductive Hypothesis**:
   - Assume $P(k)$ is true for some $k \geq 1$, that is, $1 + 2 + ... + k = \frac{k(k+1)}{2}$
4. **Inductive Step**:
   - $1 + 2 + ... + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)((k+1)+1)}{2}$
   - Therefore, $P(k+1)$ is true
5. Therefore, by PMI, $P(n)$ is true for all $n \in \mathbb{Z}^+$

Proof by Strong Mathematical Induction (2PI)

- Let $P(n)$ be a property that is defined for integers n, and let a and b be fixed integers with $a \leq b$. Then, we can prove that $P(n)$ is true for all integers $n \geq a$ as follows:

1. Show $P(a), P(a+1), ..., P(b)$ are are true
2. Show that for any integer $k \geq b$, if $P(i)$ is true for all integers i from a through k, then $P(k+1)$ is true

- Another variation of strong induction is to show that:

1. $P(a)$ is true
2. For every $k \geq a$, $(P(a) \wedge P(a+1) \wedge ... \wedge P(k)) \Rightarrow P(k+1)$

Example 1

Prove for all integers $n \geq 12$, $n = 4a + 5b$ for some $a, b \in \mathbb{N}$

Proof by 2PI:
1. Let $P(n) \equiv (n = 4a + 5b)$, for some $a, b \in \mathbb{N}, n \geq 12$
2. **Basis Step**: Show that $P(12), P(13), P(14), P(15)$ hold.
   1. (Steps Omitted)
3. Assume $P(i)$ holds for $12 \leq i \leq k$, given some $k \geq 15$
4. **Inductive Step**: To show $P(k+1)$ is true
   1. $P(k-3)$ holds (by induction hypothesis), so $k - 3 = 4a + 5b$ for some $a, b \in \mathbb{N}$
   2. $k + 1 = (k-3) + 4 = (4a + 5b) + 4 = 4(a+1) + 5b$
   3. Hence, $P(k+1)$ is true
5. Therefore, $P(n)$ is true for $n \geq 12$

Example 2

Prove that any integer greater than 1 is divisible by a prime number

Proof by 2PI:
1. Let $P(n) \equiv (n$ is divisible by a prime$)$, for $n > 1$
2. **Basis Step**: $P(2)$ is true since 2 is divisible by 2
3. **Inductive Step**: To show that for some $k \geq 2$, if $P(i)$ is true for all integers i from 2 through k, then $P(k+1)$ is also true
   1. Case 1 ($k+1$ is prime): In this case $k+1$ is divisible by a prime number which is itself
   2. Case 2 ($k+1$ is not prime): In this case, $k + 1 = ab$ where a and b are integers with $1 < a < k+1$ and $1 < b < k+1$
      1. Thus, in particular, $2 \leq a \leq k$, and by the inductive hypothesis, a is divisible by a prime number p
      2. Furthermore, since $k + 1 = ab$, $k+1$ is divisible by a
      3. By **transitivity of divisibility**, $k+1$ is divisible by a prime p
4. Therefore, any integer greater than 1 is divisible by a prime

Other Conjecture, Proof and Disproof

- **Fermat's last theorem** (*Proved*)

$$\exists x, y, z \in \mathbb{Z}, n \geq 3 \to x^n + y^n = z^n$$

  - `"I have discovered a truly remarkable PROOF of this theorem which this margin is too small to contain, Fermat"`
  - Fermat's last theorem (Conjecture) was not rigorously proven, until it was done so, by transforming it into Taniyama-Shimura Conjecture, which is then proved by Andrew Wiles in 1995.
- **Goldbach Conjecture** (Not yet proven)

$$\forall k \in \text{even integer}, k = m + n \text{ where m and n are prime numbers}$$

- **Euler Conjecture** (Disproved)

$$\forall a, b, c, d \in \mathbb{R}, a^4 + b^4 + c^4 = d^4 \text{ has nontrivial whole number solution}$$
$$\text{disproved with } 95800^4 + 217518^4 + 414560^4 = 422481^4$$

1.6 Elementary Number Theory

Integers: Even and Odd Parity

Let $n \in \mathbb{Z}$

Predicate	Definition
Even(n)	$\exists k \in \mathbb{Z}, n = 2k$
Odd(n)	$\exists k \in \mathbb{Z}, n = 2k + 1$

Theorem 4.1.1 Sum of Two Even Integers is Even (p 169)

Prove: If two integers are even, then their sum is even. That is,

$$\forall x, y \in \mathbb{Z}(\text{Even}(x) \wedge \text{Even}(y) \Rightarrow \text{Even}(x+y))$$

Proof:
1. Let $m, n \in \mathbb{Z}$ be two particular but arbitrarily chosen integers
2. Suppose m is even and n is even (antecedent)
   1. Then, $\exists j \in \mathbb{Z}$ such that $m = 2j$ (by definition of even integers)
   2. Then, $\exists k \in \mathbb{Z}$ such that $n = 2k$ (by definition of even integers)
   2. $m + n = 2j + 2k = 2(j+k)$ (by basic algebra)
   3. $j + k \in Z$ (by closure of integers under subtraction)
   4. Hence $m + n$ is even (by definition of even integers)
3. Therefore if two integers are even, their sum is even. `[Q.E.D]`

Integers: Prime and Composite Numbers

Let $n \in \mathbb{Z}$

Predicate	Definition
Prime(n)	$\forall r, s \in \mathbb{Z}, (n > 1) \wedge (n = rs)$ $\to (r = 1 \wedge s = n) \vee (r = n \wedge s = 1)$
Composite(n)	$\exists r, s \in \mathbb{Z}, (n > 1) \wedge (n = rs)$ $\to 1 < r < n$

Rational Numbers

Symbol	Definition
$r \in \mathbb{Q}$	$\exists a \in \mathbb{Z}, b \in \mathbb{Z}, r = \frac{a}{b} \wedge b \neq 0$

Example: Repeating digits

Prove: 0.1212121212… is rational number.

Proof:
1. Let $x = 0.12121212...$, Then $100x = 12.1212121...$
2. Thus $100x - x = 12.121212... - 0.12121212 = 12$
3. But also $100x - x = 99x$
4. Hence, $99x = 12$
5. and so $x = \frac{12}{99}$
6. $\therefore 0.121212... = \frac{12}{99}$ which is ratio of two nonzero integers and thus, is a rational number.

Theorem 4.3.1: Every integer is a rational number (p185)

Prove: $\forall x \in \mathbb{R} x \in \mathbb{Z} \to x \in \mathbb{Q}$

Proof:
1. Assume $x \in \mathbb{Z}$
2. $x = \frac{x}{1}$
3. $\frac{x}{1} \in \mathbb{Q}$ as $x \in \mathbb{Z} \wedge 1 \in \mathbb{Z} \wedge 1 \neq 0$ and thus by definition, x is Rational

Theorem 4.3.2: The sum of any two rational numbers is rational (p186)

Divisibility

Symbol	Definition
d divides n, $d \mid n$	$\exists k \in \mathbb{Z}, n = kd \wedge d \neq 0$

The following statements are equivalent:
- n is divisible by d
- d is factor of n
- d is divisor of n
- d divides n

Absolute Number

Symbol	Definition
absolute value of x, $\lvert x \rvert$	$\lvert x \rvert = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$

Floor and Ceiling

Symbol	Definition
floor x, $\lfloor x \rfloor$	$\lfloor x \rfloor = n \Leftrightarrow n \leq x < n + 1$
ceiling x, $\lceil x \rceil$	$\lceil x \rceil = n \Leftrightarrow n - 1 < x \leq n$

2 Relations

2.1 Set Theory

$$A = \{x \in S. \mid P(x)\}$$

- A, S are sets.
- x is element of S.
- $\emptyset \notin S$
- $P(x)$ is property that element S may or may not suffice.

Subset, Proper Subset, Equal Set

- **Subset** $\subseteq$

$$A \subseteq B \Leftrightarrow$$
$$\forall x, x \in A \Rightarrow x \in B$$

- **Not Subset (Negation)** $\nsubseteq$

$$A \nsubseteq B \Leftrightarrow$$
$$\exists x, x \in A \Rightarrow x \notin B$$

- **Proper Subset** $\subsetneq$

$$A \subsetneq B \Leftrightarrow$$
$$A \subseteq B \wedge \exists x, x \in B \Rightarrow x \notin A$$

- **Equal Set =**

$$A = B \Leftrightarrow$$
$$A \subseteq B \wedge B \subseteq A$$

Example: Proof of Subset

$$A = \{m \in \mathbb{Z} \mid m = 6r + 12 \text{ for some } r \in \mathbb{Z}\}$$
$$B = \{n \in \mathbb{Z} \mid n = 3s \text{ for some } s \in \mathbb{Z}\}$$

Proof that $A \subseteq B$
1. Suppose $x \in \mathbb{Z}$, and $x \in A$
2. By definition of A, there is integer r, that $x = 6r + 12$
3. Let $s = 2r + 4$
4. s is an integer because product and sums of integers are integers.
5. Also $3s = 3(2r + 4) = 6r + 12 = x$
6. $\therefore$ By definition of B, x is an element of B

Relations among Sets of Number

$$\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

Sets Operations

Let U be the **universal set**.

- **Union** of A and B, $(A \cup B)$

$$A \cup B = \{x \in U \mid x \in A \vee x \in B\}$$

- **Intersection** of A and B, $(A \cap B)$

$$A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$$

- **Difference** of B minus A, $(B - A)$

$$B - A = \{x \in U \mid x \in B \wedge x \notin A\} = B \cap \bar{A}$$

- **Complement** of A $(\bar{A})$

$$\bar{A} = \{x \in U \mid x \notin A\}$$

Empty Set, Disjoint Set, Power Set

- **Empty Set** $(\emptyset, \{\})$

$$\emptyset = \{\forall x \in U, x \in \emptyset\}$$

- **Disjoint Sets**, A and B are disjoint

$$A \cap B = \emptyset$$

- **Power Sets** $(\mathcal{P}(A))$ is set of all subsets of A

$$\mathcal{P}(\{x, y\}) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$$
$$\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B) \text{ , Tut 5, 8(i)}$$
$$\mathcal{P}(A \cup B) \nsubseteq \mathcal{P}(A) \cup \mathcal{P}(B) \text{ , Tut 5, 8(ii)}$$

Subset Relations

No.	Law	Identities
1	Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
2	Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
3	Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4	Identity laws	$A \cap U = A$ $A \cap \emptyset = \emptyset$ $A \cup U = U$ $A \cup \emptyset = A$
5	Complement laws	$\bar{\bar{A}} = A$ $A \cap \bar{A} = \emptyset$ $\bar{U} = \emptyset$
6	Idempotent laws	$A \cup A = A$ $A \cap A = A$
7	De Morgan's laws	$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$
8	Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

Set Identities

No.	Law	Identities
1	Inclusion of Intersection	$A \cap B \subseteq A$ $A \cap B \subseteq B$
2	Inclusion in Union	$A \subseteq A \cup B$ $B \subseteq A \cup B$
3	Transitive Property of Subsets	$A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$
4	Tutorial 3, Q5 (iii)	$A \subseteq B \rightarrow A \cap C \subseteq B \cap C$
5	Tutorial 3, Q5 (iv)	$B \subseteq A \leftrightarrow A \cap B = B$

2.2 Relations

2.2.1 Ordered Pair, Cartesian Product, Relation

Ordered Pair, Cartesian Product

- **Ordered Pair**: (a, b) consist of the element a (first element) and b (second element)

$$(a, b) = (c, d) \leftrightarrow (a = c) \wedge (b = d)$$

- **Cartesian Product** of set $A_1, A_2, ..., A_n$ denoted $A_1 \times A_2 \times ... \times A_n$, is set of ordered n-tuples $(a_1, a_2, ..., a_n)$ where $a_1 \in A_1, a_2 \in A_2, ..., a_n \in A_n$

$$A_1 \times A_2 \times ... \times A_n = \{(a_1, a_2, ..., a_n) \mid a_1 \in A_1, a_2 \in A_2, ..., a_n \in A_n\}$$

Example: Cartesian Products

Let $A = \{x, y\}, B = \{1, 2, 3\}, C = \{a, b\}$

$$A \times B = \{(x, 1), (y, 1), (x, 2), (y, 2), (x, 3), (y, 3)\}$$
$$B \times A = \{(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)\}$$
$$A^2 = A \times A = \{(x, x), (x, y), (y, x), (y, y)\}$$
$$(A \times B) \times C = \{(u, v) \mid u \in A \times B \wedge v \in C\} = \{((x, 1), a), ((x, 2), a), ...\}$$
$$A \times B \times C = \{(u, v, w) \mid u \in A \wedge v \in B \wedge w \in C\}$$
$$\emptyset \times A = \{(u, v) \mid u \in \emptyset \wedge v \in A\} = \emptyset$$

Relation and Arrow Diagram

- **Relation, R from A and B** $\subseteq A \times B$.

Given ordered pair $(x, y) \in A \times B$

$$xRy \leftrightarrow (x, y) \in R$$
$$x \not R y \leftrightarrow (x, y) \notin R$$

Where set A is **domain** of R, set B is **co-domain**

- **Arrow Diagram** describes a relation from Domain to Codomain, where

$$x \rightarrow y \Leftrightarrow xRy \Leftrightarrow (x, y) \in R$$

Composite and Inverse Relation

- **Composite Relation, $R' \cdot R$** where R and R' be binary relations that map X to Y and Y to Z.

$$R' \cdot R = R'(R(x)) = \{(x, z) \in X \times Z \mid \exists y \in Y, (x, y) \in R \wedge (y, z) \in R'\}$$

- **Inverse Relation, R^{-1}**

$$\{(y, x) \in Y \times X \mid (x, y) \in R\}$$
$$\forall x \in X, \forall y \in X, (y, x) \in R^{-1} \Leftrightarrow (x, y) \in R$$

- **Inverse Composite Relation $(R' \cdot R)^{-1}$**

$$(S \cdot R)^{-1} = R^{-1} \cdot S^{-1}$$

Example: Prove for Composition is Associative for Relation

Let A, B, C, D be sets and $R \subseteq A \times B, S \subseteq B \times C, T \subseteq C \times D$

Prove $T \cdot (S \cdot R) = (T \cdot S) \cdot R$

$$T \cdot (S \cdot R) = (T \cdot S) \cdot R \Leftrightarrow$$
$$T \cdot (S \cdot R) \subseteq (T \cdot S) \cdot R \wedge (T \cdot S) \cdot R \subseteq T \cdot (S \cdot R)$$

Case 1: $T \cdot (S \cdot R) \subseteq (T \cdot S) \cdot R$

$$\text{Suppose } (a, d) \in T \cdot (S \cdot R)$$
$$\exists c \in C \text{ such that } (a, c) \in S \cdot R \wedge (c, d) \in T$$
$$\exists b \in B \text{ such that } (a, b) \in R \wedge (b, c) \in S \wedge (c, d) \in T$$
$$\text{Therefore } (a, b) \in R \wedge (b, d) \in T \cdot S, \therefore (a, d) \in (T \cdot S) \cdot R$$
$$T \cdot (S \cdot R) \subseteq (T \cdot S) \cdot R$$

Case 2: $(T \cdot S) \cdot R \subseteq T \cdot (S \cdot R)$

$$\text{Suppose } (a, d) \in (T \cdot S) \cdot R$$
$$\exists b \in B \text{ such that } (a, b) \in R \wedge (b, d) \in T \cdot S$$
$$\exists c \in C \text{ such that } (a, b) \in R \wedge (b, c) \in S \wedge (c, d) \in T$$
$$\text{Therefore } (a, c) \in S \cdot R \wedge (c, d) \in T, \therefore (a, d) \in (S \cdot R) \cdot T$$
$$(T \cdot S) \cdot R \subseteq T \cdot (S \cdot R)$$
$$\therefore (T \cdot S) \cdot R = T \cdot (S \cdot R)$$

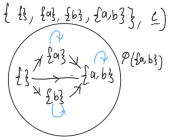
2.2.2 Graph

Directed Graph

- **Directed Graph** is a 2-tuple (V, D) where:
  - V is an **non-empty set**, and $v \in V$ v is a node / vertices
  - D is a **binary relation** on V, and $(x, y) \in D$ is an edge from x to y
  - edge (x, x) is also called a **loop**.

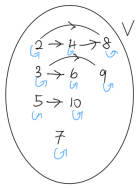
Example

$(\mathcal{P}(\{a, b\}), \subseteq) =$



Example

$V = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $D = \{(m, n) \in V \times V \mid m \text{ divides } n\}$



- **Undirected Graph** is a 2-tuple (V, E) where:
  - V is an **non-empty set**, and $v \in V$ v is a node / vertices
  - E is set of elements of form $\{u\}, \{u, v\}$ where $u, v \in V$ and $u \neq v$. Element of E is called an edge.
  - edge $\{u\} \in E$ is a loop

2.2.3 Equivalence Relation and Classes

Reflexive, Symmetric, Transitive, Equivalence

- **Reflexive** binary relation R on X (Each element is related to itself)

$$R \text{ is reflexive} \Leftrightarrow \forall x \in X, xRx \Leftrightarrow \forall (x_1, x_2) \in X \times X, (x_1, x_2) \in R$$
$$i.e. \text{ Check } (x, x) \in R / \forall x \in X, xRx$$

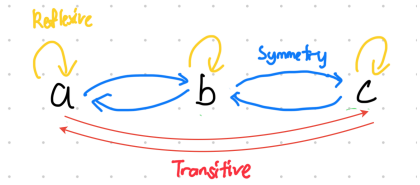
- **Symmetric** binary relation R on X (Each element has back-and-forth connection)

$$R \text{ is symmetric} \Leftrightarrow \text{if } (x, y) \in R \rightarrow (y, x) \in R$$
$$i.e. \text{ Check } (x, y) \in R \wedge (y, x) \in R / xRy \rightarrow yRx$$

- **Transitive** binary relation R on X (There is always direct connection connecting two pair of relation)

$$R \text{ is transitive} \Leftrightarrow \forall xRz, xRy \wedge yRx$$
$$\Leftrightarrow R \cdot R \subseteq R$$
$$i.e. \text{ Check } yRz \cdot xRy \rightarrow xRz$$

- **Equivalence Relation** is relation that is reflexive, symmetric and transitive.
  - i.e. groups things that look different but actually the same
  - rational number are really just equivalence classes
    - $(a, b)R(c, d) \Leftrightarrow ad = bc; \therefore \frac{1}{2} = \frac{-1}{-2} = \frac{2}{4} = \frac{3}{6} = ...$



Example of Equivalence Relation

Let $R_4 = \{(m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid 4 \text{ divides } m - n\}$

Proof of Reflexive

$$xR_4x$$
$$\equiv x - x = k4$$
$$\equiv 0 = k4$$
$$\text{Since } \exists k = 0 \in \mathbb{Z} \text{ where } x - x = k(4), \forall x \in \mathbb{Z}^+,$$
$$\therefore R_4 \text{ is reflexive}$$

Proof of Symmetric

$$\text{Let } x \in \mathbb{Z}^+, y \in \mathbb{Z}^+$$
$$xR_4y$$
$$\equiv (x - y) = k4$$
$$\equiv (x - y) = (-k)4$$
$$\equiv (y - x) = k4$$
$$\equiv yR_4x$$
$$\therefore R_4 \text{ is symmetric}$$

Proof of Transitive

$$\text{Let } x \in \mathbb{Z}^+, y \in \mathbb{Z}^+ y \in \mathbb{Z}^+$$
$$xR_4y \wedge yR_4z$$
$$\equiv \exists k \in \mathbb{Z}x - y = k4 \wedge \exists h \in \mathbb{Z}y - z = h4$$
$$\equiv \exists k \in \mathbb{Z} \wedge h \in \mathbb{Z}, x - (z + h4) = k4$$
$$\equiv \exists k \in \mathbb{Z} \wedge h \in \mathbb{Z}, x - z = 4(k - h)$$
$$\equiv \exists p \in \mathbb{Z}, x - z = 4p$$
$$\equiv xR_4z$$
$$\therefore R_4 \text{ is transitive}$$

Proof of Equivalence

R_4 is equivalence as it is reflexive, transitive and symmetric

Partition, Equivalence Class

- **Partition, A** of X is a set in form of $\{A_1, A_2, ...\}$ where all the following conditions are satisfied:

1. $A_1, A_2, ...$ be non-empty subsets of X
2. $A_1 \cup A_2 \cup ... = X$
3. $A_i \cap A_j = \emptyset$, for $A_i \neq A_j$ ("aka pairwise disjoint")

- **Equivalence Class [a]** of a under R, equivalence relation on X, $a \in X$ and

$$[a]_R = \{x \in X \mid xRa\}$$
$$\forall x \in X, x \in [a] \Leftrightarrow xRa \Leftrightarrow (x, a) \in R$$

- Lemma 8.3.2 (pg 513)
  - Suppose R is equivalence relation on A (*If a is related to be, a and b belongs to same equivalence class*)

$$aRb \Rightarrow [a] = [b]$$

- Lemma 8.3.3 (pg 514) (*Equivalence class can only be either the same or disjoint*)

$$U = [a] \cup [b] = \emptyset \vee [a] = [b]$$

- **Relation Induced by Partition, Π** is a partition

$$\forall (x, y) \in A_i, xRy$$
$$\equiv \Pi = \{[b]_R \in \mathcal{P}(X) \mid b \in X\}$$

2.2.4 Partial Order Relations

Anti-Symmetry, Partial Order, Total Order

- **Anti-Symmetry** binary relation R on X

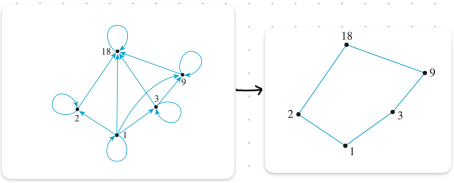
$$R \text{ is anti-symmetry} \Leftrightarrow xRy \wedge yRx \rightarrow x = y$$
$$i.e. \text{ No reverse line, only loop}$$

- **Partial Order Relation** is relation that is reflexive, antisymmetric and transitive.

Hasse Diagram

- Named after Helmet Hasse.
- Given any **partial order relation** defined on a **finite set**, it is possible to draw **directed graph** in a way that the following properties are satisfied / eliminated
  1. Remove all loops at all the vertices
  2. Remove all arrows whose existence is implied by transitive property
  3. Remove direction indicators on the arrows

$a|b \Leftrightarrow b = ka$ for some $k \in \mathbb{Z}$



Partial Order Relation → Hasse Diagram

- **Total Order Relation, R**

$$\Leftrightarrow (R \text{ is partial order relation}) \wedge \forall a, b \in A, aRb \vee bRa$$

  - Hasse diagram for total order relation forms a chain.

Well-Ordering Principles for Integers (WOP)

- Y is **bounded below (aka lowerbound)** by b $\Leftrightarrow b \leq x \forall x \in Y$. Let $Y \subseteq \mathbb{R} \wedge b \in \mathbb{R}$
- Suppose $S \subseteq \mathbb{Z}, S \neq \emptyset$ and S is bounded below by an integer. Then S contains a smallest integer.

Topological Sorting

Let $\preceq$ be partial order relation on a nonempty finite set A, Topological Sort sorts element into a chain given the relation.

1. Pick any minimal element $x \in A$
2. Set A' = A - {x}
3. While $A' \neq \emptyset$
   1. Pick minimal element $y \in A$
   2. Append $x \preccurlyeq y$
   2. $A' := A' - \{y\}$

Example:
- Let A = {2,3,4, 6, 18, 24}, Relation is divide relation |
- Total Order: $3 \preceq 2 \preceq 6 \preceq 18 \preceq 4 \preceq 24$

2.3 Function

- **Function** from set X to set Y, denoted $f : X \rightarrow Y$ is a relation with **domain X** and **co-domain Y** that satisfies the following two properties:
  1. Every element in X is related to Y

$$\forall x \in X, \exists y \in Y \text{ such that } (x, y) \in F$$

  2. no element in X is related to more than one element in Y, {one, many}-to-one

$$(x, y) \in F \wedge (x, z) \in F \rightarrow y = z$$

- **Range of f** or **Image of X under Y**

$$\{y \in Y \mid y = f(x) \text{ for some } x \in X\}$$
$$\text{Range} \subseteq \text{Codomain}$$

- **Inverse image of y** or **Preimage of y**

$$\{x \in X \mid f(x) = y\} \text{ Inverse Image} \subseteq \text{Domain}$$

Boolean Function

$$f : \{T, F\}^n \rightarrow \{T, F\}$$

Identity Function

$$i_X : X \rightarrow X$$
$$i_X = f^{-1} \circ f$$
$$f \circ i_X = f$$

Equality of Function

Let $f : X \rightarrow Y$ and $g : X \rightarrow Y$,

$$f = g \Leftrightarrow \forall x \in X, f(x) = g(x)$$

Composition of Function

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$,

$$g \circ f : X \rightarrow Z \Leftrightarrow$$
$$\forall x \in X, (g \circ f)(x) \equiv g(f(x))$$

- Corollary 2.9

$$f : X \rightarrow Y$$
$$(i_Y \circ f = f) \wedge (f \circ i_X = f)$$

2.3.1 One-to-One, Onto, and Inverse Functions

One-to-One Function (Injective)

$$F \text{ is one-to-one} \Leftrightarrow F(x_1) = F(x_2) \rightarrow x_1 = x_2 \Leftrightarrow x_1 \neq x_2 \rightarrow F(x_1) \neq F(x_2)$$
$$F \text{ is not one-to-one} \Leftrightarrow \exists x_1, x_2 \in X, F(x_1) = F(x_2) \wedge x_1 \neq x_2$$

Aka same output imply same input

Example of Composite 1-1 Proof

Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are both one-to-one function, shows that $g \circ f$ is one-to-one function.

$$(g \circ f)(x) = (g \circ f)(x')$$
$$g(f(x)) = g(f(x'))$$
$$\text{Since g is one-to-one } f(x) = f(x')$$
$$\text{Since f is one-to-one } x = x' \text{ \#QED}$$

Onto Function (Surjective)

$$F \text{ is onto} \Leftrightarrow \forall y \in Y, \exists x \in X, F(x) = y$$
$$F \text{ is not onto} \Leftrightarrow \exists y \in Y, \forall x \in X, F(x) \neq y$$

Example of Composite onto Proof

Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are both onto function, shows that $g \circ f$ is onto function.

$$\text{Let } z \text{ be any } Z,$$
$$\text{Since g is onto}, \forall z \in Z, \exists y \in Y, z = g(y)$$
$$\text{Since f is onto}, \exists x \in X, f(x) = y$$
$$\therefore \exists x \in X \text{ such that } (g \circ f)(x) = g(f(x)) = g(y) = z \text{ \#QED}$$

Aka every output has its input

Bijective Function

Function that is both one-to-one and onto.

- A bijective function always has bijective inverse
- Composition of bijective functions is bijective

Inverse Function

Let F be bijective function, then there exists $F^{-1} : Y \rightarrow X$

$$F^{-1}(y) = x \Leftrightarrow y = F(X)$$

Example of Cardinality of Equivalence Classes

Define $\underset{k}{\equiv}$ be equivalence relation such that $\{(m, n) \in \mathbb{R}^2 \mid k \text{ divides m-n}\}$. Proof that its equivalence classes have same cardinality.

Let $[b]_{\underset{k}{\equiv}}$ and $[c]_{\underset{k}{\equiv}}$ be equivalence classes.

To prove that both classes have same cardinality, there exists **bijective** function,

$$f : [b]_{\underset{k}{\equiv}} \rightarrow [c]_{\underset{k}{\equiv}}$$

Let $x \in \mathbb{R}$, by definition,

$$[b]_{\underset{k}{\equiv}} = \{x \in \mathbb{R} \mid k \text{ divides } x - b\}$$
$$= \{x \in \mathbb{R} \mid \exists z \in \mathbb{Z}, x - b = zk\} = \{x \in \mathbb{R} \mid \exists z \in \mathbb{Z}, x = zk + b\}$$

Let $y \in \mathbb{R}$, by definition,

$$[c]_{\underset{k}{\equiv}} = \{y \in \mathbb{R} \mid k \text{ divides } x - c\}$$
$$= \{y \in \mathbb{R} \mid \exists z \in \mathbb{Z}, y - c = zk\} = \{y \in \mathbb{R} \mid \exists z \in \mathbb{Z}, y = zk + c\}$$

Therefore, f is defined as follows,

$$f(x) = y$$
$$f(zk + b) = zk + c$$
$$(zk + b) - b + c = zk + c$$
$$f(x) = x - b + c$$

Prove that f is 1-1:

$$f(x) = f(x') \Rightarrow x - b + c = x' - b + c \Rightarrow x = x' \text{ \#QED}$$

Prove that f is onto:

$$\text{For any } y \in [c]_{\underset{k}{\equiv}}$$
$$y = f(x) = x - b + c$$
$$x = y + b - c \text{ (define x based on y)}$$
$$f(x) = (y + b - c) - b + c = y \text{ (demonstrate f(x) = y)}$$
$$\therefore \exists x \in [b]_{\underset{k}{\equiv}}, \forall y \in [c]_{\underset{k}{\equiv}}, f(x) = y \text{ \#QED}$$

Thus, f is bijective and so $[b]$ and $[c]$ has same cardinality.

3 Counting

3.1 Countability and Un-countability

Cardinality and Equivalence

- **Cardinality** refers to the size of a set, $|A| = n$. (i.e. what is the maximum n such that I can list all the elements one-by-one exhaustively)
- For any A and B be any (finite/infinite) sets, A and B has **same cardinality**

$$\Leftrightarrow \exists \text{ bijection } f : A \xrightarrow[\text{onto}]{\text{1-1}} B$$

- Set A is **finite set**

$$\Leftrightarrow (A = \emptyset) \vee (\exists \text{ bijection } f : \{1, 2, ..., n\} \in \mathbb{Z}^+ \rightarrow A)$$
$$\text{Size A} = |A| = n$$

- Set B is **countably infinite**

$$\Leftrightarrow A \neq \emptyset \wedge \text{same cardinality as } \mathbb{Z}^+$$
$$\text{OR (relaxed condition) } \exists \text{ surjection } f : \mathbb{N} \xrightarrow{\text{onto}} A$$

- Set is **countable**

$$\Leftrightarrow \text{Set is finite} \vee \text{Set is countably infinite}$$
$$\text{else, set is uncountable}$$

Theorem 3.1: Pigeonhole Principle (pg 605)

Suppose $n, k \in \mathbb{Z}^+, X = \{x_1, x_2, ..., x_n\}, Y = \{y_1, ..., y_k\}$
1. If $n > k$, there is no 1-1 function $f : X \rightarrow Y$
2. If $n < k$ there is no onto function $g : X \rightarrow Y$
3. $n = k \Leftrightarrow$ there is a bijection $h : X \rightarrow Y$

Proof:
1. - Proof by contradiction. Suppose $n > k$ and there exists 1-1 function $f : X \rightarrow Y$
   - For any $x, x' \in X$, suppose $x \neq x'$, there is no guarantee that $F(x) = F(x')$ as there must be a box with two or more domain
   - Thus, this contradicts with the definition of 1-1 function
   - $\therefore$ by contradiction, there is no 1-1 function $f : X \rightarrow Y$
2. - Suppose $n < k$, by definition of range of function, range$(f) \subseteq$ codomain
   - so $|\text{range}(g)| \leq |X| = n < k = |Y|$
   - Since $|\text{range}(g)| < |Y|$, range$(y) \subsetneq Y$, and therefore $\exists y \in Y, \forall x \in X, f(x) \neq y$ and function g is not onto
3. 1. $\Rightarrow$ Suppose n=k, define $h : X \rightarrow Y$ such that $H(x_i) = y_i \forall i$
      - Show 1-1. Suppose $x_i, x_j \in X$ and $x_i \neq x_j, H(x_i) = y_i \neq y_j = H(x_j)$
      - Show Onto. Suppose for any $y_i \in Y, x_i \in X, F(x_i) = y_i$
      - $\therefore \exists$ bijection $h : X \rightarrow Y$
   2. $\leq$ Suppose exists bijection $h : X \rightarrow Y$
      - By properties of bijection:
        - h is 1-1, By contraposition of 3.1.1 $n \leq k$
        - h is onto, by contraposition of 3.1.2 $n \geq k$

   $\therefore n \leq k \wedge n \geq k \Rightarrow n = k$

Properties of Cardinality

Cardinality is an **equivalence relation**. Therefore, for any two sets A and B, its cardinality is:
- **reflexive**: A has same cardinality as A,
  - $|A| = |A|$
- **symmetric**: If A has same cardinality as B, then B has the same cardinality as A
  - $|A| = |B| \rightarrow |B| = |A|$
- **transitive**: If A has same cardinality as B and B has same cardinality as C, A has same cardinality as C,)
  - $(|A| = |B|) \wedge (|B| = |C|) \rightarrow |A| = |C|$

$\mathbb{Z}^+$,$\mathbb{N}$ and $\mathbb{Z}$ has same cardinality

$$\text{Even though } \mathbb{Z}^+ \subsetneq \mathbb{N} \subsetneq \mathbb{Z},$$
$$|\mathbb{Z}^+| = |\mathbb{N}| = |\mathbb{Z}| = \text{countably infinite}$$

Theorem 3.2: $\mathbb{N}$ and $\mathbb{Z}$ has same cardinality

For $\mathbb{N}$ and $\mathbb{Z}$ to has same cardinality, we need to show that $\exists$ bijection $F : \mathbb{N} \rightarrow \mathbb{Z}$

Define:

$$F(n) = \begin{cases} \frac{n}{2} \mid \text{n is even} \\ -\frac{n+1}{2} \mid \text{n is odd} \end{cases}$$
$$F(0) = 0, F(1) = 1, F(2) = -1, F(3) = 2, F(4) = -2, ...$$

Show that f is 1-1:
- Suppose $f(b) = f(c) = k$, for $b, c \in \mathbb{N}$
  - Case k is even:

$2f(b) = 2(\frac{b}{2}) = b = c = 2(\frac{c}{2}) = 2f(c)$

- Case k is odd:

$1 - 2f(b) = 1 - 2\frac{-b+1}{2} = b = c = 1 - 2(-c + \frac{1}{2}) = f(c)$

- $\therefore f(b) = f(c) \Rightarrow b = c$

Show that f is onto:

- For any $k \in \mathbb{Z}$:
  - Case $k \geq 0$
    - Let $x = 2k$, $f(x) = f(2k) = 2\frac{k}{2} = k$
    - $\exists x = 2k, \forall k \in \mathbb{Z}$ for some $k \geq 0, f(x) = k$
  - Case $k < 0$
    - Let $x = -2k - 1$, $f(x) = \frac{-x+1}{2} = k$
    - $\exists x = -2k - 1, \forall k \in \mathbb{Z}$ for some k < 0, $f(x) = k$

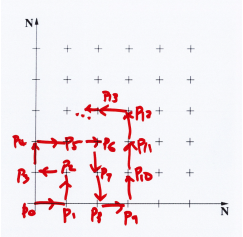
$\therefore f$ is onto

Since, $f : \mathbb{N} \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} \mathbb{Z}$, $\therefore$ $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality. #QED

Theorem 3.3: $\mathbb{N}\times\mathbb{N}$ is countable

For $\mathbb{N}\times\mathbb{N}$ is countable, there exists $f : N \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} \mathbb{N}\times\mathbb{N}$.

Define $f(n) = p_n$ where p_n refers to element in first quadrant of (x,y)-plane.



Show that f is 1-1:

- Suppose $f(j) = f(k) = p$, since square walk visits every point exactly once, p is therefore only visited once
- $\therefore f(j) = f(k) \Rightarrow j = k$, f is 1-1

Show that f is onto:

- Consider any $(x, y) \in \mathbb{N}^2$,
- Point (x, y) will be eventually visited in the k-th step.
- $\therefore (x, y) = p_k = f(k)$, f is onto

Since, $f : \mathbb{N} \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} \mathbb{N}\times\mathbb{N}$, $\therefore$ $\mathbb{N}\times\mathbb{N}$ is countable. #QED

Theorem 3.4: $\{0, 1\}^*$ is countable

For $\{0, 1\}^*$ is countable, there exists $f : N \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} \{0, 1\}^*$.

$f(0) = \emptyset, f(1) = 1, f(2) = 10, f(3) = 11, ...$

- $|\mathbb{Q}|$, Exists bijection $G : \mathbb{Z}^+ \rightarrow \mathbb{Q}^+$ where G list the numbers diagonally.

FIGURE 7.4.3

$$g(1) = \frac{1}{2}, g(2) = \frac{1}{2}, g(3) = \frac{2}{1}, g(4) = \frac{3}{1}, g(5) = \frac{1}{3}\left(\text{skip } \frac{2}{2}\right)...$$

Countability of Subset for Countable (Infinite) Set

Theorem 3.5: Subset of countable set is countable (pg 480)

Let A be countable set and $B \subseteq A$, Proof that A is countable $\rightarrow B$ is countable

For B to be countable set, B is either:

1. Case B is finite set:
   - By definition, a finite set is countable set.
   - $\therefore$ B is countable set that is subset of A, another countable set.
2. Case B is infinite set:
   - Since B is infinite set, A is also infinite set.
   - Since A is countable, elements of A can be enumerated $A = \{A_1, A_2, ...\}$
   - Since $B \subseteq A$, $B = \left\{A_{i_0}, A_{i_1}, ...\right\}$ for some $i_0 < i_1$.
   - Define $f : \mathbb{N} \rightarrow B, f(n) = A_{i_n}$
   - Show that f is 1-1:
     - Let $b, c \in \mathbb{N}$. Suppose $b \neq c$,

$$b < c \text{ (Since b != c)}$$
$$\Rightarrow i_b < i_c$$
$$\Rightarrow A_b < A_c \text{ (Given that X is countable)}$$
$$\Rightarrow f(b) \neq f(c)$$

     - $\therefore$ By contraposition $f(b) = f(c) \Rightarrow b = c$ (#QED)
   - Show that f is onto:
     - For any $b \in B$.
     - Then $y = A_{i_k}$ for some k (By definition of subset)
     - $y = A_{i_k} = f(k)$

     $\therefore \exists k \in \mathbb{N}, \forall b \in B, f(k) = b$
   - Since $f : \mathbb{N} \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} B$, $\therefore$ B is countable set.

Theorem 3.6: subset of infinite set is countable

Let X be infinite set and $A \subseteq X$. Show that A is countably infinite.

- For A to be countably infinite, A is countable.
- For A to be countable, exists bijection $f : \mathbb{N} \rightarrow A$
- Define f as follows:
  - n := 0
  - A := $\emptyset$
  - repeat:
    - Choose any y = X - A
    - a_n := y
    - A := $A \cup \{a_n\}$
    - n := n+1
  - end
- Proof f is 1-1
  - Suppose $n_1 \neq n_2$
  - $n_1 < n_2$
  - $a_{n_2} \in X - \left\{a_0, ..., a_{n_1}, ..., a_{n_2 - 1}\right\}$
  - $a_{n_1} \neq a_{n_2}$
  - $f(n_1) \neq f(n_2)$
- Proof f is onto
  - For any $y \in A$
  - $y = a_k$ for some k
  - $y = f(x)$

$\therefore$ f is bijective and A is countably infinite

Uncountable Set

Theorem 3.7: A countably infinite set has uncountable many subset

i.e. X is countably infinite set. $\Rightarrow \mathcal{P}(X)$ is uncountable.

Proof by Cantor Diagonalization:

- Let X be countably infinite, and therefore, $f : \mathbb{N} \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} X, f(i) = x_i$.
- $(\{x_0\}, \{x_1\}, \{x_2\}) \subseteq X, \in \mathcal{P}(X)$
- **Suppose $\mathcal{P}(X)$ is countably infinite**, there exists bijection $g : \mathbb{N} \underset{\text{onto}}{\overset{1\text{-}1}{\rightarrow}} \mathcal{P}(X))$
- Since f is bijection, $f^{-1} : X \rightarrow \mathbb{N}$ is bijection
- Since f^{-1} and g is bijection, $g \circ f^{-1} : X \rightarrow \mathcal{P}(X)$ is bijection
- Define bijection $h = g \circ f^{-1}$ and h maps an element to a particular subset.
- Arranging $\{x_0, x_1, ...\}\times\{h(x_0), h(x_1), ...\}$ in a 2D-grid, define

$$e_{ij} = \begin{cases} 1 \text{ if } x_i \in h(x_j) \\ 0 \text{ if } x_i \notin h(x_j) \end{cases}$$

- Define diagonal set $D = \{x_i \in X \mid e_{ii} = 0\} = \{x_i \in X \mid x_i \notin h(x_i)\}$
- Since h is onto, $\exists x_k \in X$ such that $h(x_k) = D$
- $x_k \in D$?
  - Suppose $x_k \in D$ $e_{kk} = 0 \Rightarrow x_k\neg \in h(x_k) \Rightarrow x_k\neg \in D$ (contradiction)
  - Suppose $x_k\neg \in D$ $x_k\neg \in h(x_k) \Rightarrow e_{kk} = 0 \Rightarrow x_k \in D$ (contradiction)
- This contradiction proves that $\mathcal{P}(X)$ is **not countably infinite**.

- $\mathbb{R}$ is uncountable (Cantor Diagonalization Theorem)
  - See textbook pg480 for detailed proof. But tldr, there always exists new number that is not in the list of decimal numbers. Therefore, set of all number 0 and 1 is uncountable.
- Any set with uncountable subset is uncountable, Corollary 7.4.4 (pg 481)

3.2 Counting of Disjoint Sets

Let A, B be finite, disjoint set. (i.e. $A \cap B = \emptyset$)

- **Addition Rule**, Lemma 3.9.1

$$|A \cup B| = |A| + |B| \in \mathbb{R}$$

- **Difference Rule**, when $B \subseteq A$ Lemma 3.9.2

$$|A - B| = |A| - |B|$$

- **Union(s) of finite sets is finite**, Theorem 3.10
- **Inclusion / Exclusion Rule for 2 and 3 sets**, Theorem 3.11

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Useful set manipulation identities:

$$X \cap (Y - X) = X \cap (Y \cap X') = (X \cap X') \cap Y = \emptyset \cap Y = \emptyset$$
$$X \cup (Y - X) = X \cup (Y \cap X') = (X \cup Y) \cap (X \cup X') = (X \cup Y) \cap U = X \cup Y$$

Proof that $A \cup B$ is countable

Let A and B be countable and disjoint set.

For $A \cup B$ be countable set, $A \cup B$ are either finite or countably infinite.

Case A is finite and B is finite.

- Then $A \cup B$ are finite, as $|A \cup B| = |A| + |B|$ by addition rule
- $\therefore A \cup B$ is countable.

Case A is finite and B is countably infinite.

- If $A = \emptyset$
  - $A \cup B = \emptyset \cup B = B$, which is countable
- If $A \neq \emptyset$
  - A can be arranged as $A = \{a_0, ..., a_{k-1}\}$
  - Since B is countable, there exist bijection $f : \mathbb{N} \rightarrow B$
  - Define $g : \mathbb{N} \rightarrow A \cup B$ where

$$g(n) = \begin{cases} a_k \mid n < k \\ f(n - k) \mid n \geq k \end{cases}$$

  - g is 1-1
    - Suppose $n \neq n'$
      - Case $n < k$ and $n' < k$: $g(n) = a_k \neq a_{k'} = g(n')$
      - Case $n < k$ and $n' \geq k$: $g(n) \in A, g(n') \in B$
        - Since $A \cap B = \emptyset, g(n) \neq g(n')$
      - Case $n \geq k$ and $n' \geq k$: $g(n) = f(n) \neq f(n') = g(n')$
  - g in onto
    - Let $x \in A \cup B$
      - If $x \in A$, $x = a_n = g(n)$ for some $0 \leq n < k$
      - If $x \in B$, $x = f(m)$, $x = g(m + k)$

Case A and B is countably infinite

- There exists bijections $f : \mathbb{N} \rightarrow A, g : \mathbb{N} \rightarrow B$
- Define $h \rightarrow \mathbb{Z} \rightarrow A \cup B$ where

$$h(n) = \begin{cases} f(n) \mid n \geq 0 \\ g(n) \mid n < 0 \end{cases}$$

- h is 1-1
  - Suppose $n \neq n'$
    - Case $n < 0$ and $n' < 0$: $g(n) = h(n) \neq h(n') = g(n')$
    - Case $n \geq 0$ and $n' \geq 0$: $g(n) = f(n) \neq f(n') = g(n')$
    - Case $n < 0$ and $n' \geq 0$: $g(n) \in A, g(n') \in B$
      - Since $A \cap B = \emptyset, g(n) \neq g(n')$
- h is onto
  - Let $x \in A \cup B$
  - If $x \in A$, $x \in f(n)$ for some $x \in \mathbb{N}$, $x = h(n)$
  - If $x \in B$, $x \in g(m)$ for some $m \in \mathbb{N}$, let n = (-m+1)
    - Then $n < 0$, so $h(n) = g(-n - 1) = g(m) = x$
- Since h is bijective, $A \cup B$ has same cardinality as $\mathbb{Z}$ which is countable, so $A \cup B$ is also countable.

Multiplication Rule

- **Multiplication Rule**

$$|A\times B| = |A||B|$$

Theorem 3.12: $|A_1\times A_2\times...\times A_n| = |A_1||A_2|...|A_n|$ for any $n \geq 2$

By 1PI on n

Basis:

- $|A_1\times A_2| =$

  Let $n = 2$, $|A_1|\ |A_2|$ (by multiplication rule)

Induction Hypothesis:

- Assume claim is true for $n = k$ for some $k \geq 2$

Induction Step:

- Consider case $n = k + 1$
- Then

$$|A_1\times...\times A_k\times A_{k+1}| = |\{(a_1, a_2, ..., a_k, a_{k+1} \mid a_1 \in A_1 \wedge ...a_{k+1} \in A_{k+1})\}|$$
$$= |\{(a_1, a_2, ..., a_k, a_{k+1} \mid a_i \in A_i\}|$$
$$= |D\times A_{k+1}| \text{ Let } D = A_1\times...\times A_k$$
$$= |D|\ |A_{k+1}| \text{ by multiplication rule}$$
$$= |A_1|\ ...\ |A_k|\ |A_{k+1}| \text{ by inductive hypothesis}$$

- $\therefore$ claim is true for $n = k + 1$

By 1PI, the claim is true for $n \geq 2$

- $A = A_1 = ... = A_n \Rightarrow |A^n| = |A_1\times...\times A_n| = |A|^n$

3.3 Combinations

- r-combination of s

$$nCr = \frac{n!}{r!(n - r)!}$$

- r-permutation of s

$$nPr = \frac{n!}{(n - r)!}$$

- Tutorial 10, qn1

$$nCr + nC(r - 1) = (n + 1)Cr$$

- Binomial Theorem, Tutorial 10, qn3

$$(x + y)^n = nC0x^ny^0 + nC1x^{n-1}y^1 + ... + nCrx^{n-r}y^r + ...nCnx^0y^n$$

4 Graph

4.1 Graph, Paths, Cycles

Let G be **undirected graph** $G = (V, E)$, where V is nonempty set of **vertices**, and E is **edges**.

Let G' be **directed graph**, $G' = (V, E)$ consists of two finite sets, where V is nonempty set of vertices and D of directed edge, $e = (u, w)$. e is therefore the directed edge from v to w.

Degree of vertices, $\deg(v)$ equals number of edges that are incident on v, with an edge that is a loop counted twice.

Countability of Graph

- G is **trivial** $\Leftrightarrow |V| = 1$ (i.e. only 1 vertices)
- G is **finite** if V is **finite**
- G is **infinite** if V is **infinite**

Subgraph, Proper Subgraph

H is **subgraph** of G

$$\Leftrightarrow (V_H \subseteq V_G) \wedge (E_H \subseteq E_G)$$

H is **proper subgraph** of G

$$\Leftrightarrow (H \text{ is subgraph of } G) \wedge (H \neq G)$$

Path, Connectedness, Cycle

Let x_1, x_p be **vertices** in graph G and $p \in \mathbb{N} \geq 2$. A **path between x_1 and x_p in G** is a subgraph defined as:

$$\text{path} = (\{x_1, ..., x_p\}, \{\{x_1, x_2\}, \{x_2, x_3\}\}, ..., \{x_{p-1}, x_p\}),$$
$$|\text{path}| = p - 1$$

An undirected graph G is **connected**

$$\Leftrightarrow (G \text{ is trivial}) \vee (\forall \text{ vertices } v, w \in V_G, \exists \text{ path}, p \text{ from } v \text{ to } w)$$

Edge as a Relation

Let R **be a binary relation**, describe an edge connecting two distinct vertices.

$$R = \{(b, c) \in V \times V \mid b \neq c \wedge \{b, c\} \in E\}$$

R_n **is defined as:**, for $n \in \mathbb{Z}^+$

$$R_n = \begin{cases} R \mid n = 1 \\ R \circ R_{n-1} \mid n \geq 2 \end{cases}$$

R_+ **transitive closure** is defined as (aka fking long chain)

$$R_+ = \cup_{n=1}^{\infty} R_n$$

Theorem 4.1.1: There is path of length n between x and y, then $(x, y) \in R_n$

Proof by 1PI on n:

Base Case: Let $n = 1$.

- Suppose there is a path of length n between x and y, by definition of path, there exists subgraph $G_1 = (\{x, y\}, \{\{x, y\}\})$.
- Since $x \neq y$ and $(x, y) \in E_{G_1}$, $(x, y) \in R$ by definition of relation

Induction Hypothesis: Suppose claim is true for $n = k \geq 1$, then $(x_1, x_k) \in R_k$

Induction Step: Let n = k+1

- Suppose there is a path of length $k + 1$ between x_1 and x_{k+1}, by definition of path, there exists subgraph $G_{k+1} = (\{x_1, ..., x_k, x_{k+1}\}, \{\{x_1, x_2\}, ..., \{x_k, x_{k+1}\}\})$
- Since $x_k \neq x_{k+1}$ and $\{x_k, x_{k+1}\} \in E_{G_{k+1}}$, $(x_k, x_{k+1}) \in R$
- By I.H., $(x_1, x_k) \in R_k$, and $(x_k, x_{k+1}) \in R$, $(x_1, x_{k+1}) \in R \circ R_k = R_{k+1}$ #QED

Theorem 4.1.2: If $(x, y) \in R_n$, there is path of length at most n between x and y

Proof by 1PI on n

Base Case: Let $n = 1$.

- Suppose $(x, y) \in R_1$, $x \neq y$ and $(x, y) \in E_{G_1}$, $(x, y) \in R$ by definition of relation.
- Since there exists subgraph $G_1 = (\{x, y\}, \{\{x, y\}\})$, there is a path of length 1 between x and y.

Induction Hypothesis: Suppose claim is true for $n = k \geq 1$, then there is path of length at most k between x and y

Induction Step: Let n = k+1

- Suppose $(x, z) \in R_{k+1} = R \circ R_k = yRz \wedge xR_ky$
- For $(y, z) \in R$, there is an edge of length 1 between y and z.
- By I.H., since xR_ky, there is a path of length at most k between x and y
- $\therefore$ length of path between x and z = length of path between x and y + length of path between y and $z \leq k + 1$ #QED

Corollary 4.2: G is connected $\Leftrightarrow \forall x, y \in V, x \neq y \Rightarrow (x, y) \in R_+$

$\Rightarrow$ Suppose G is connected.

- By definition of connected, for any $x, y \in V$, there exists a path from x to y for length n.
- By theorem 4.1.1, $(x, y) \in R_n$ for some n
- By definition of transitive closure, $(x, y) \in R_+$ since $n \in \mathbb{Z}$.

$<=$ Suppose $\forall x, y \in V, x \neq y \Rightarrow (x, y) \in R_+$

- $R_+ = \cup_{n=1}^{\infty} R_n$.
- By theorem 4.1.2, there is a path between x and y of at most length n, for any $x, y \in V$
- By definition of connected graph, G is connected.

Loop, Cycle, Cyclic and Acyclic

A **cycle** exists in undirected graph with $|V| \geq 3$

$$\Leftrightarrow \exists \text{ path } p = (\{x_1, x_2, ..., x_n\}, \{\{x_1, x_2\}, \{x_2, x_3\}, ..., \{x_n, x_1\})$$

An undirected graph is **cyclic** if it consists a loop or cycle.

An undirected graph is **acyclic** if no cycle is found.

Theorem 4.3: Let G be undirected graph with **no loop**, G is **cyclic**

$$\Leftrightarrow \exists u, v \in V \text{ with more than 1 path between them}$$

Connected Component

H be **connected component** of G $\Leftrightarrow$

1. H is subgraph of G
2. H is connected
3. no connected subgraph of G has H as a subgraph and contains vertices or edges that are not in H.

(Saitama's official definition:) G does not contain another connected subgraph H', such that H is a proper subgraph of H'

(Theorem 4.4) Let x, y be distinct node in G.

Theorem 4.4: There is a path in G between x and $y \Leftrightarrow x$ and y are in the same connected component in G

$<=$ Suppose x and y are in the same connected component H of G.

- By definition of connected component, H is connected
- By definition of connected, there is a path between x and y in H
- Since H is a subgraph of G, there is a path between x and y in G

=> By contraposition, x and y are not in same connected component => there is no path in G between x and y

- To prove by contradiction, suppose x and y are not in same connected component, we need to show there is a path in G between x and y.
- Since x and y are not in the same connected component, x and y are not connected
- By definition of connected, there is no path between x and y, which contradicts with the claim that there is a path in G between x and y.
- By contradiction, x and y are not in same connected component => there is no path in G between x and y

Corollary 4.5: x and y are in the same equivalence class under R $\Leftrightarrow x$ and y are in the same connected component in G

Let A be nonempty set, R an equivalence relation on A, G be undirected graph representing R

=> Suppose x and y are in the **same equivalence class** under R.

- $x \in [b]_R \wedge y \in [b]_R$
- xRb, yRb
- Since R is symmetric, $yRb \Rightarrow bRy$
- Since R is transitive, $xRb \wedge bRy \Rightarrow xRy$, $(x, y) \in R$
- By theorem 4.1.1, there exists path between x and y in G.
- By theorem 4.4, x and y are in the same connected component

$<=$ Suppose x and y are in the **same connected component** in G.

- There is a path between x and y in G of length n for some n
- By theorem 4.1.1, $(x, y) \in R_n$
- $xRy = x \in [y]_R$
- Since R is equivalence class, R is reflexive, yRy, $y \in [y]_R$
- Since $x \in [y]_R \wedge y \in [y]_R$, x and y are in the same equivalence class.

4.2 Tree

Tree is a connected, acyclic, undirected graph.

Forest is an acyclic graph (that might not be connected)

(Theorem 4.6) **Tree Theorem**: the following statement on an undirected graph is equivalent:

1. G is a tree
2. removing any edge disconnect G
3. $|E| = |V| - 1$

Spanning Tree

Spanning Tree, $T(V, H)$, is a subgraph of $G(V, E)$, $H \subseteq E$, that is a tree, and consist of all nodes in G.

(Theorem 4.7) Every finite connected undirected graph has a spanning tree

Rooted Tree

A **rooted tree, T** is tree with distinguished node called root, r.

Level of node x is number of (unique) path from r to x

- Level(r) = 0

Height is maximum level of any node in T.

For any x and any y, $x \neq y$ and y in path between x and y:

- y is **ancestor** of x
- y is ancestor of $x \Rightarrow x$ is **descendant** of y

If x is descendent of p and $\text{level}(x) = \text{level}(p) + 1$

- p is called the **parent** of x
- x is called **child** of p

A node with child is called **internal node** A node with no child is called a **leaf**

Binary Tree is a rooted tree in which every node has at most 2 children.

Theorem 4.8: For any binary tree with m leaves and height h,$m \leq 2^h$

Proof by 2PI on h.

Base Case: Let h = 0.

- When height = 0, the tree only has root node, and 1 leave
- $\therefore m = 1 \leq 2^0 = 1$

Inductive Hypothesis: Assume the claim is true for all $h \leq k$ and $h > 0$

Inductive Step: Proof the claim is true for $h = k + 1$

- For h = k+1, a new root node is added on the top of the tree. There are two possible cases, for the root node:
- Case 1 Children:
  - Since new tree only has 1 children, number of leaves for new tree is the same as number of leaves for the tree with height of k
  - By I.H, $m \leq 2^k = 2^{k+1}$
- Case 2 Children:
  - Since new tree has 2 children, number of leaves for new tree is sum of number of leaves for both of the children.

$$m = m_l + m_h \leq 2^k + 2^k \leq 2^{k+1}$$

- $\therefore m \leq 2^{k+1}$
- By 2PI, claim is true for $h \geq 0$.

Theorem 4.9: Consider binary tree T, and every parent has 2 children. If T has m leaves, then it has $m - 1$ parents

Proof by 2PI on p

Base Case. Let $p = 0$

- For $p = 0$, the tree has no parent, and no children, and number of leaves 1
- $\therefore p = 0 = 1 - 1 = m - 1$

Inductive Hypothesis. Assume claim is true, for $p \leq k$ for some $k \geq 0$

Inductive Step: Proof claim is true for $p = k + 1$

- For $p = k + 1$, a new root node is added to the previous tree with exactly 2 children.
- Number of leaves, m is addition of number of leaves from left tree, m_l and number of leaves from right tree, m_r.
- Similarly, number of parents $p = p_l + p_r + 1$ where 1 is the newly added root.
- By I.H, $p_l = m_l - 1$, $p_r = m_r - 1$
- $p = p_l + p_r + 1 = (m_l - 1) + (m_r - 1) + 1 = m_l + m_r + 1 = m + 1$
- $\therefore$ claim is true for $p = k + 1$
- $\therefore$ By 2PI, claim is true for all $p \geq 0$.